



Justification trajectories for pension inequality in Chile (2016– 2023)

The role of social class and beliefs in meritocracy

Juan Carlos Castillo, Andreas Laffert, René Canales &
Tomás Urzúa

Department of Sociology, University of Chile



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Research project

Project

Team



Juan Carlos Castillo (PI)



Andreas Laffert (RA)



Tomás Urzúa (RA)



René Canales (RA)

Agenda

Peer-reviewed Articles

- Castillo, J. C., Canales Sellés, R., Laffert, A., & Urzúa, T. (2026). Justification trajectories for pension inequality in Chile (2016–2023): the role of social class and beliefs in meritocracy. *Frontiers in Sociology*, 11, 1771856. <https://doi.org/10.3389/fsoc.2026.1771856>
- Castillo, J. C., Laffert, A., Carrasco, K., & Iturra-Sanhueza, J. (2025). Perceptions of inequality and meritocracy: their interplay in shaping preferences for market justice in Chile (2016–2023). *Frontiers in Sociology*, 10, 1634219. <https://doi.org/10.3389/fsoc.2025.1634219>
- Castillo, J. C., Salgado, M., Carrasco, K., & Laffert, A. (2024). The Socialization of Meritocracy and Market Justice Preferences at School. *Societies*, 14(11), 214. doi.org/10.3390/soc14110214

Forthcoming

- Pensions' market justice and meritocracy: A structural equation modeling approach. Andreas Laffert, Tomás Urzúa, Juan Carlos Castillo & René Canales.
- Stability and comparability of meritocratic beliefs in school-age students: A measurement invariance approach across time and cohorts. Andreas Laffert, Juan Carlos Castillo, René Canales, Tomás Urzúa & Kevin Carrasco. *Submitted*.
- Inequality and deservingness in higher education in Chile. A conjoint survey experiment. Juan Carlos Castillo, Andreas Laffert, René Canales & Tomás Urzúa.

Background

The problem

1. Privatization and marketization of public goods, welfare policies, and social services (Gingrich, 2011; Streeck, 2016)
 2. Changes in the institutional architecture of the welfare state; expansion of market logics (Busemeyer & Iversen, 2020; Ferre, 2023)
 3. This economic order is reflected in a specific moral economy and policy feedback dynamics (Campbell, 2020; Fernandez & Jaime-Castillo, 2013; Mau, 2015)
- Market justice preferences (Busemeyer, 2015; Castillo et al., 2025; Koos & Sachweh, 2019; Lindh, 2015)

The Chilean case

- Economic growth alongside high inequality (Flores et al., 2020; Llorca-Jaña & Miller, 2021)
- Rigid relative mobility and strong closure at the top (López-Roldán & Fachelli, 2021; Torche, 2014)
- Deep privatization and commodification of social reproduction domains, combined with strong state dependence (Boccardo, 2020; Madariaga, 2020)
- Pensions: an **individual capitalization** system administered by **private pension funds (AFPs)**; covers ~11 million contributors, but **27% of the labor force** remains excluded due to **informality** (Superintendencia, 2025)
- Pension-related social conflict (Somma et al., 2021) and contested system legitimacy (“not with my money”) (Canales Cerón et al., 2021; Castillo et al., 2025)

Market justice preferences

- Lane (1986): market justice vs. political justice
- Normative beliefs that legitimize the idea that access to essential social services—such as healthcare, education, or pensions—should be determined by market-based criteria (Lindh, 2015, p. 895)
- Measurement: assessing whether individuals consider it fair that access to these services depends on income (Castillo et al., 2025; Kluegel et al., 1999; Lindh, 2015)

Factors associated with market justice

Individual

- Socioeconomic status—income, education, and occupation (Busemeyer & Iversen, 2020; Koos & Sachweh, 2019; Lindh, 2015; Svallfors, 2007)
- Perceptions of inequality and meritocracy (Castillo et al., 2025)
- Economic conservatism/liberalism (Lee & Stacey, 2024)

Contextual

- Social spending (Busemeyer, 2015; Busemeyer & Iversen, 2020; Immergut & Schneider, 2020)
- Economic inequality (Koos & Sachweh, 2019)
- Degree of privatization and market regulation (Koos & Sachweh, 2019; Lindh, 2015)

This study

Analyze the relationship between social class and preferences for market justice in pensions in Chile, and how these evolve over time

Examine the role of meritocracy in shaping the relationship between class and preferences for pension market justice

Hypotheses

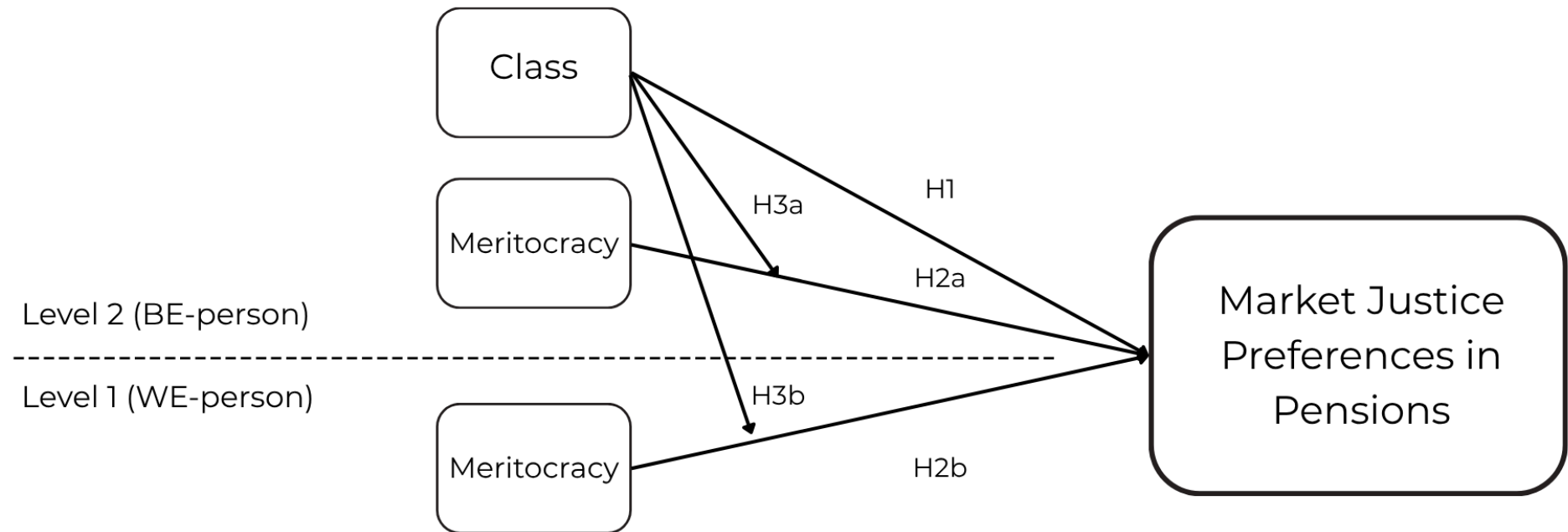


Figure 1: Hypotheses diagram

Methodology

Data

- **ELSOC (COES)**: a representative panel survey of the urban adult population in Chile, based on a probabilistic, stratified, clustered, and multistage sampling design across large and small cities
- Period: six waves (2016, 2017, 2018, 2019, 2022, and 2023)
- Attrition 2016→2023: ~ 40%
- Analytical sample (**balanced 6-wave panel**):
 - **N = 7,522 observations** (level 1) nested within
 - **N = 1,317 individuals** (level 2)

Dependent variable

- Item: ¿“It is fair that people with higher incomes have better pensions than people with lower incomes?”

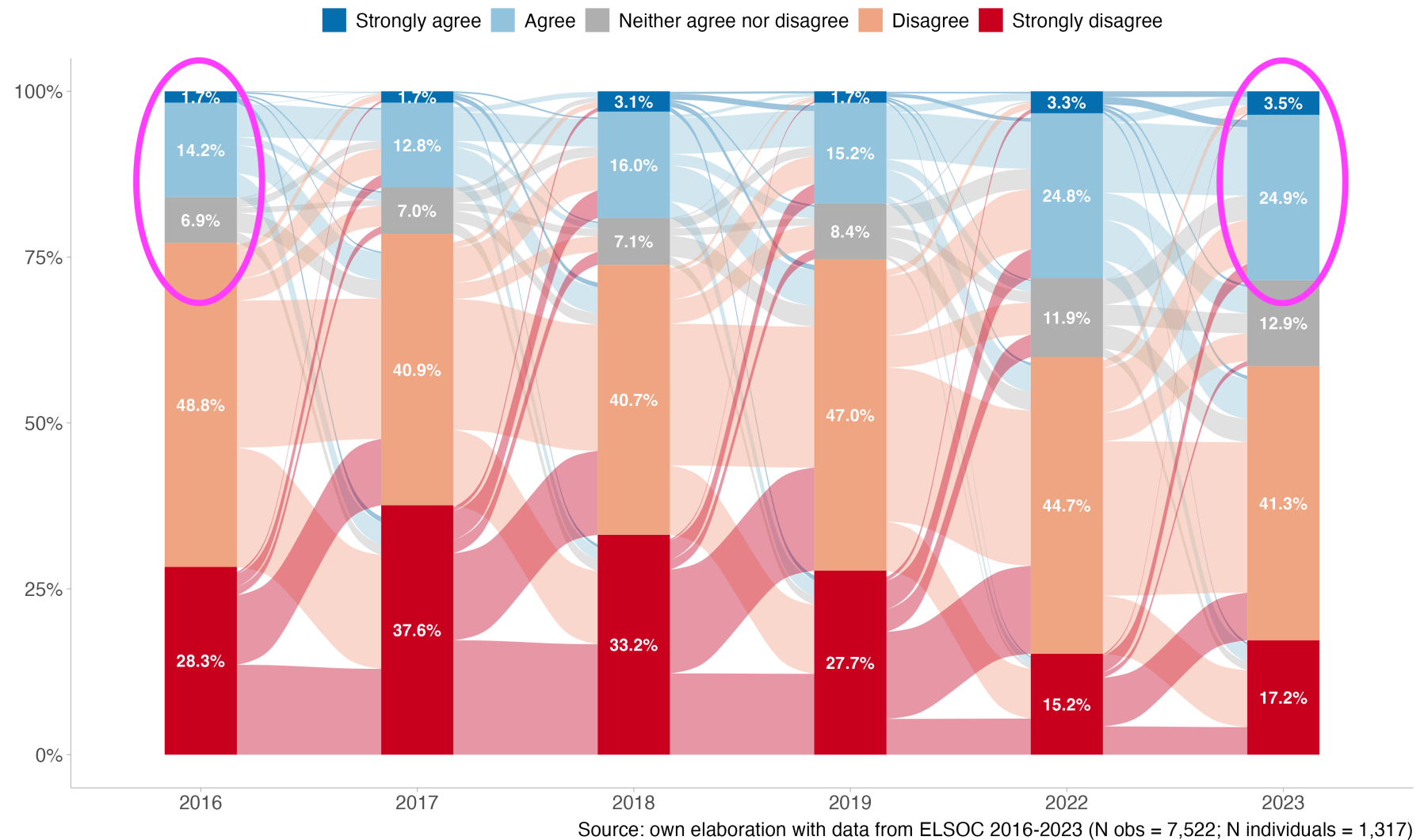
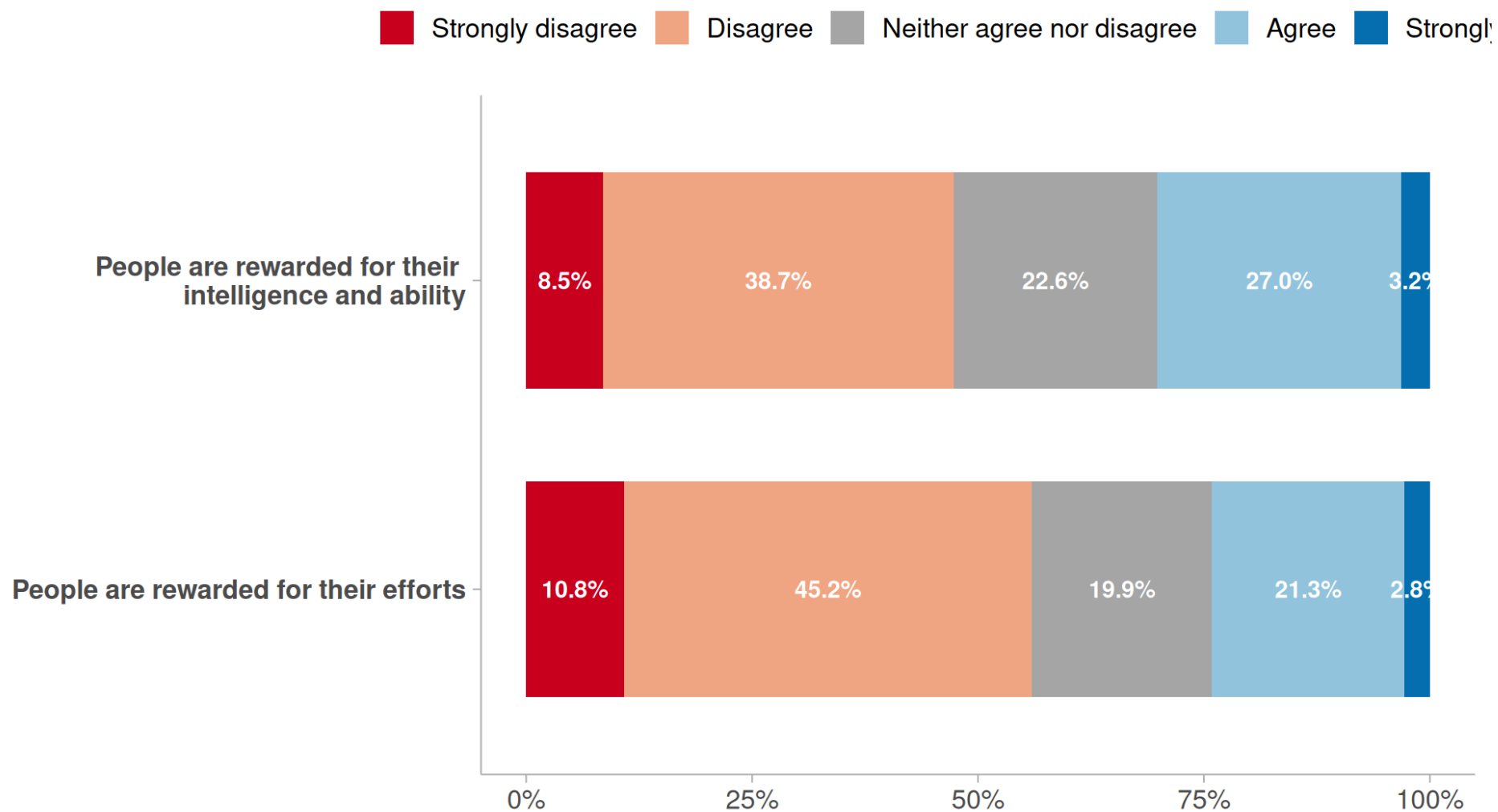


Figure 2: Distribution of responses for preferences for market justice in pensions in Chile (2016–2023)

Independent variables: Level 1



Source: own elaboration with data from ELSOC 2016-2023 (N obs = 7,522; N individuals = 1,317)

Figure 3: Distribution of responses for meritocracy items

Independent variables: Level 2

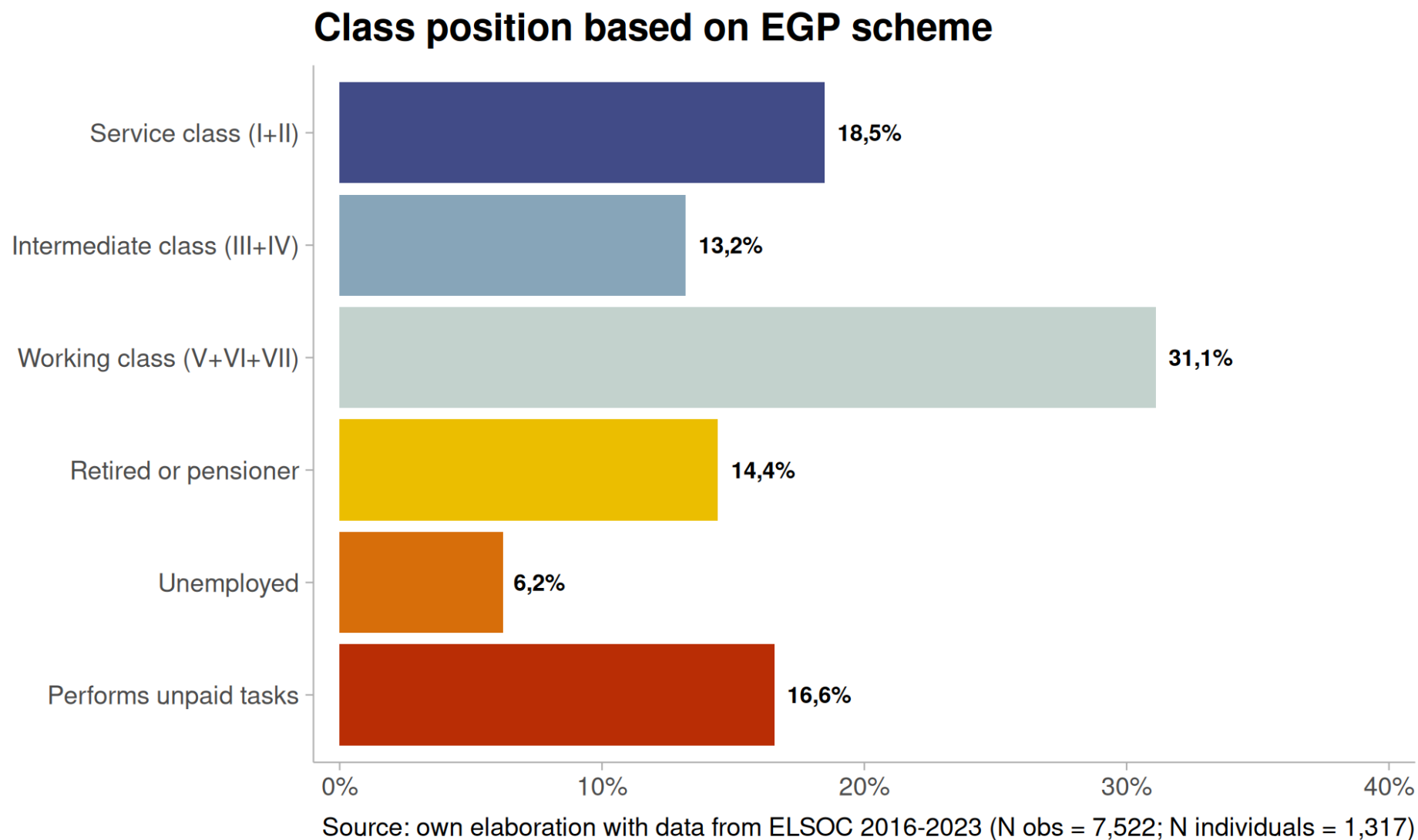


Figure 4: Frecuency of class position

- Fixed at the first wave (i.e., time-invariant) $\rightarrow$ captures average between-person differences (BE = $\bar{X}_i$)

Estimation strategy

Cumulative link mixed models (CLMM):

- Panel structure: repeated observations (level 1) nested within individuals (level 2)
- Models both *within-person* (WE) and *between-person* (BE) effects; includes random effects (intercept and time slope)
- WE/BE decomposition (person-mean centering) (Bell et al., 2019):
 - Within: $X_{it}^{WE} = X_{it} - \bar{X}_i$
 - Between: $X_i^{BE} = \bar{X}_i$
- Formally:

$$\eta_{it} = \beta_0 + \beta_1 \text{Time}_{it} + \beta_2 \text{Meritocracy}_{it}^{WE} + \beta_3 \text{Meritocracy}_i^{BE} + \beta_4 \text{Cl}$$

Results

Models

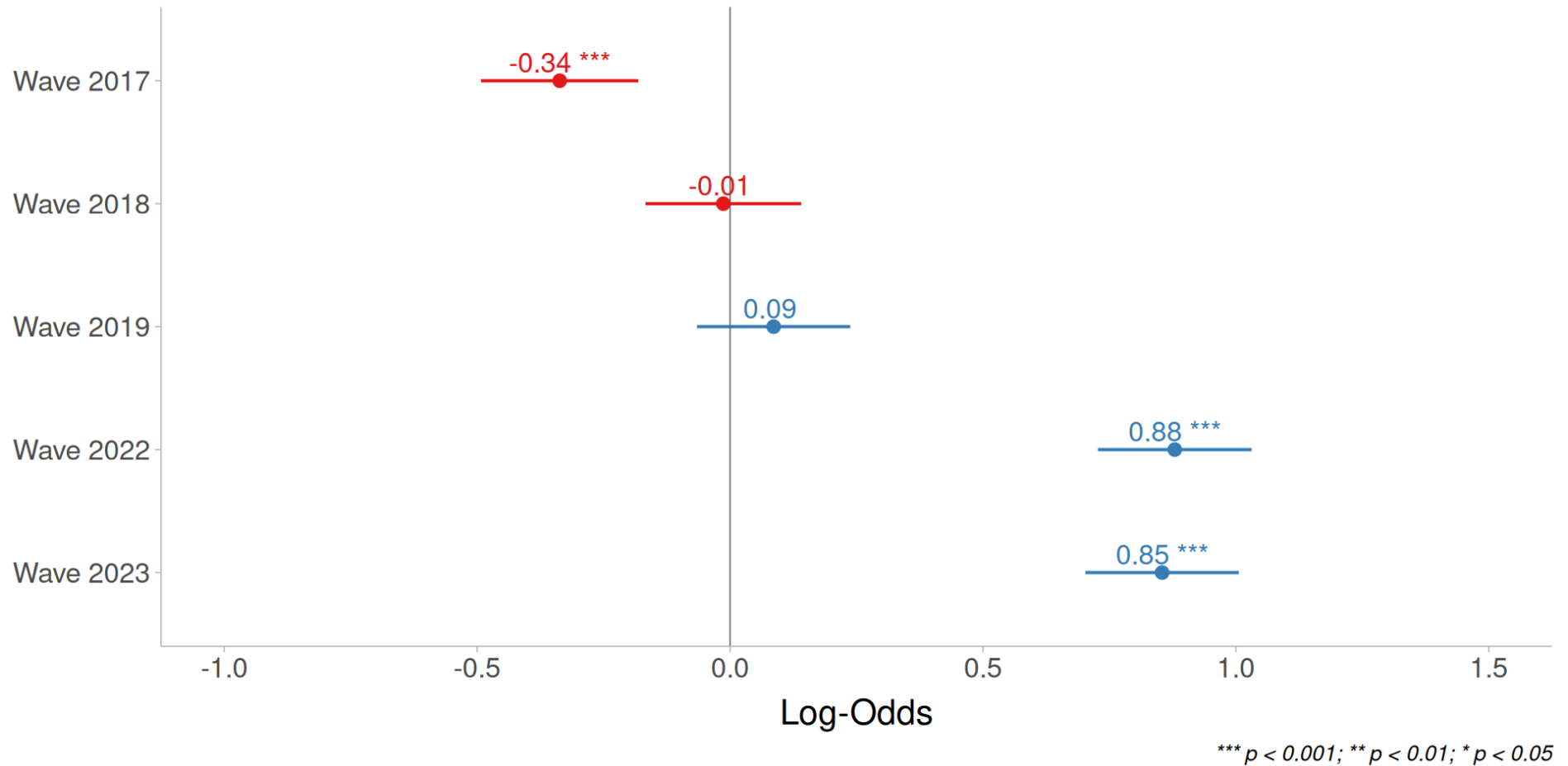


Figure 5: Changes across waves in market justice preferences in pensions

- ICC = 0.23 between, 0.77 within

Models

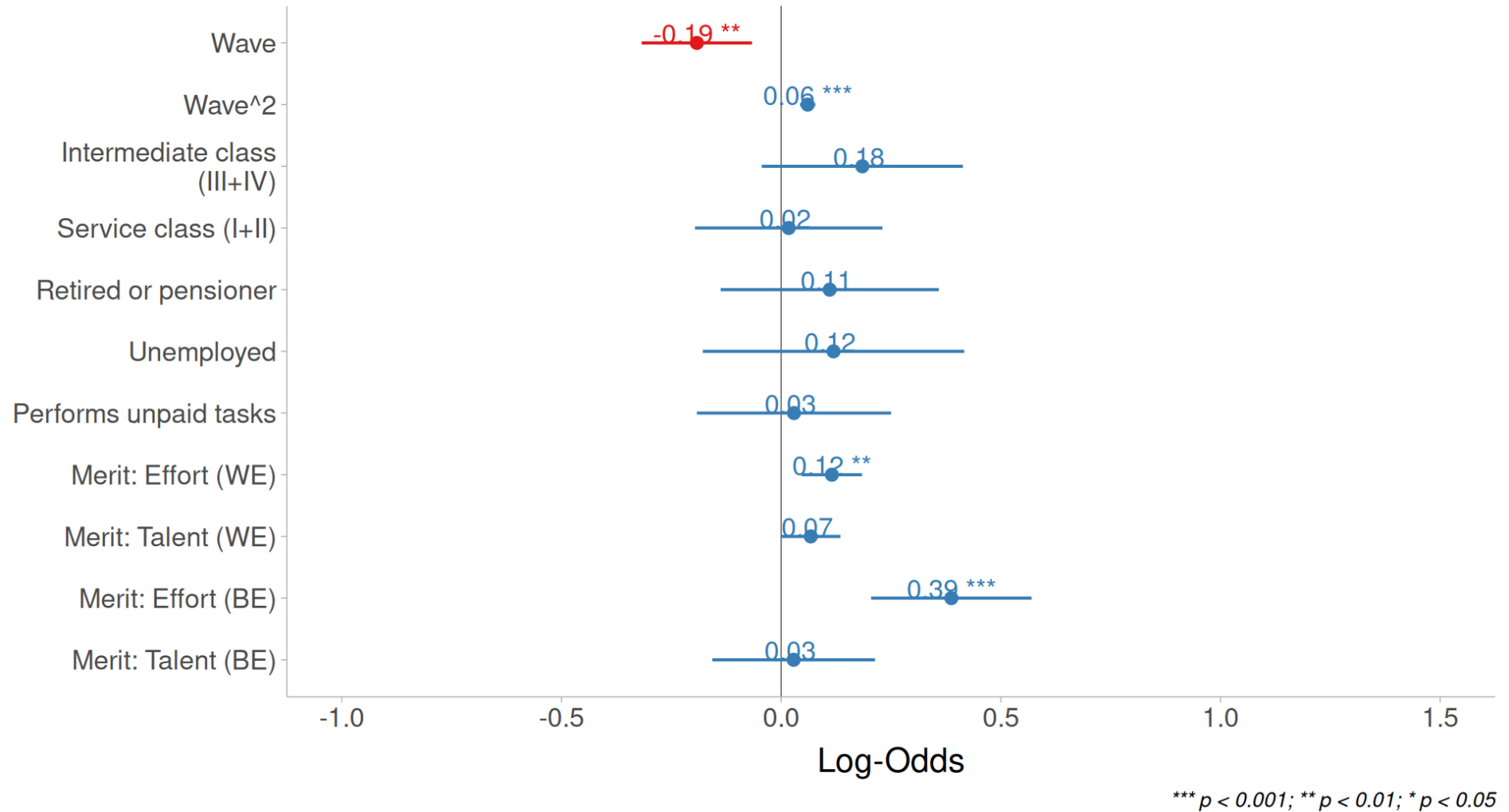


Figure 6: Main within- and between-person effects on market justice preferences in pensions

- All models control for: gender, age, educational level, and political identification

Models

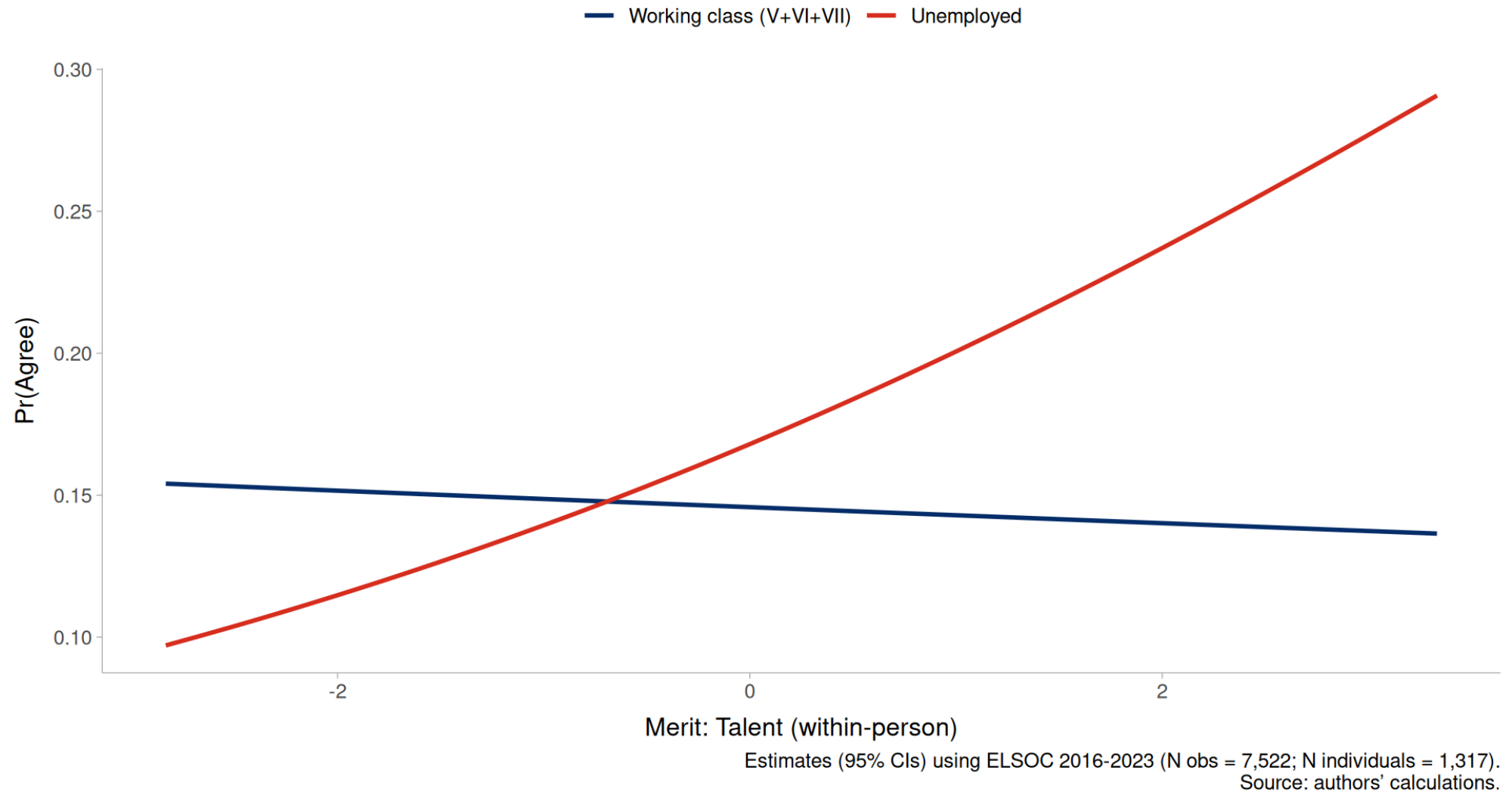


Figure 7: Predicted probabilities of pension market justice preferences by talent-based meritocratic perceptions and social class

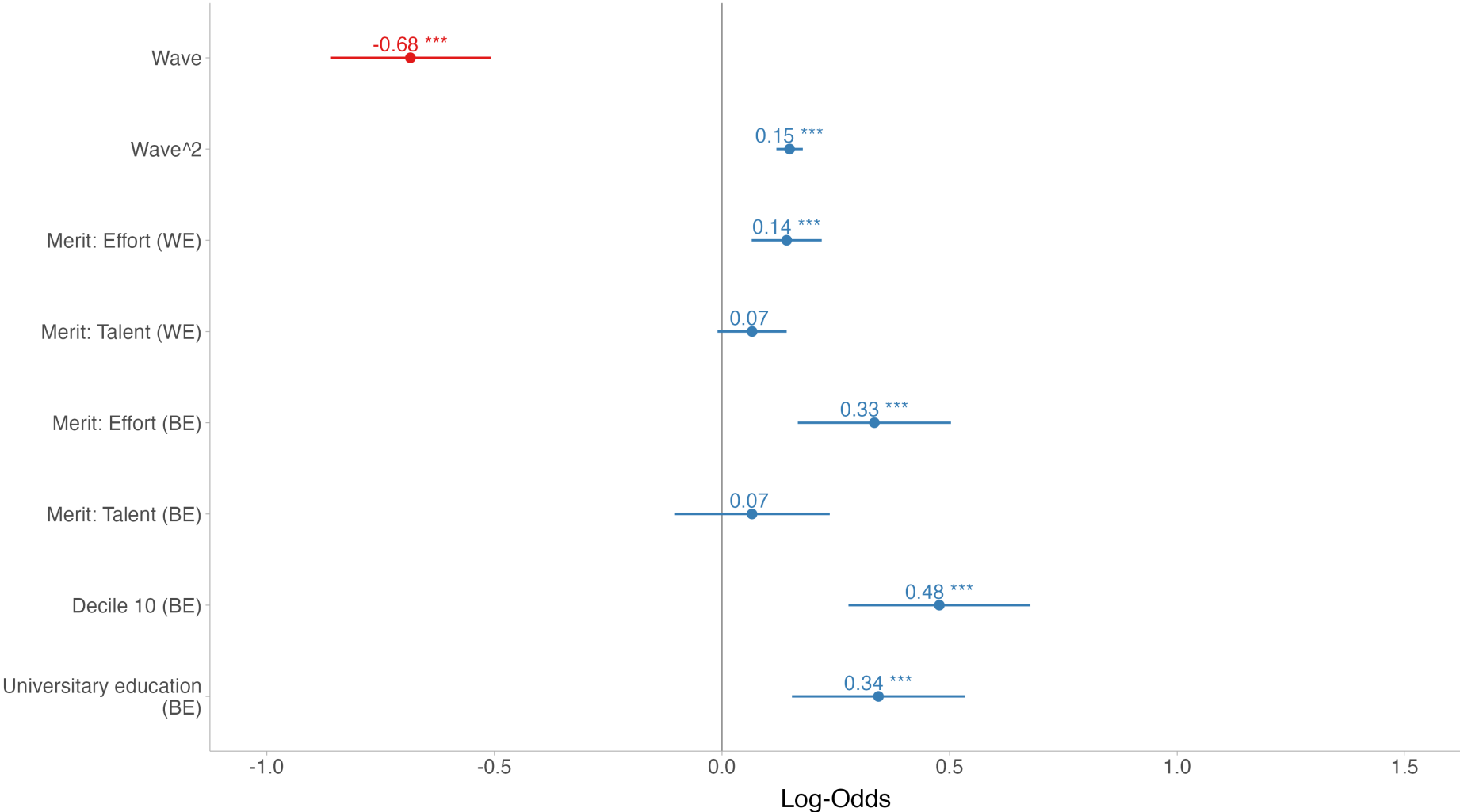
Preliminary conclusions

1. **Time:** Preferences for market justice in pensions have increased in recent years in Chile
2. **Social class:** Higher social classes don't exhibit greater support for market justice in pensions than lower classes (H1) (Busemeyer & Iversen, 2020; Kerner, 2020; Lindh, 2015)
3. **Meritocracy:** The belief that effort is rewarded is associated with higher support for market justice in pensions (H2a and H2b) (Castillo et al., 2025)
4. **Class × Meritocracy:** There is no evidence of an amplification effect (H3a/H3b)

Caveats

- No evidence of class differences in support for pension market justice, but what about a more direct measure of stratification differences?

The role of income and education



Source: own elaboration based on ELSOC data 2016-2023 (N obs = 6,123; N individuals = 1,327)
Controls: gender, age, and political identification
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Figure 8: Main within- and between-person effects on market justice preferences in pension

Caveats

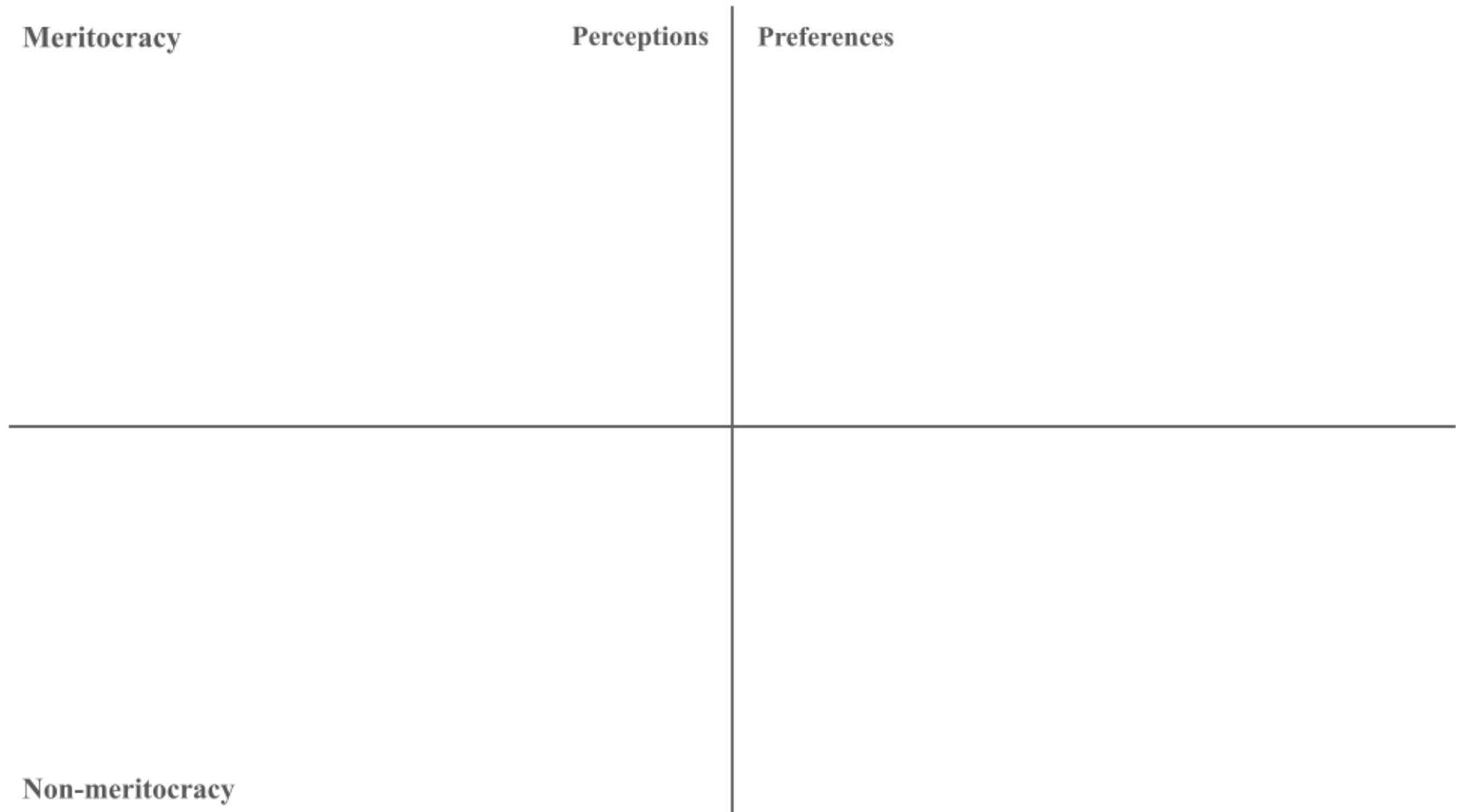
- Meritocratic beliefs show a strong and consistent effect. Yet, are measured in a highly reduced way, through effort and talent only
- Next step: model meritocracy as a multidimensional construct
→ (Castillo et al., 2023)

Additional analyses (SEM)

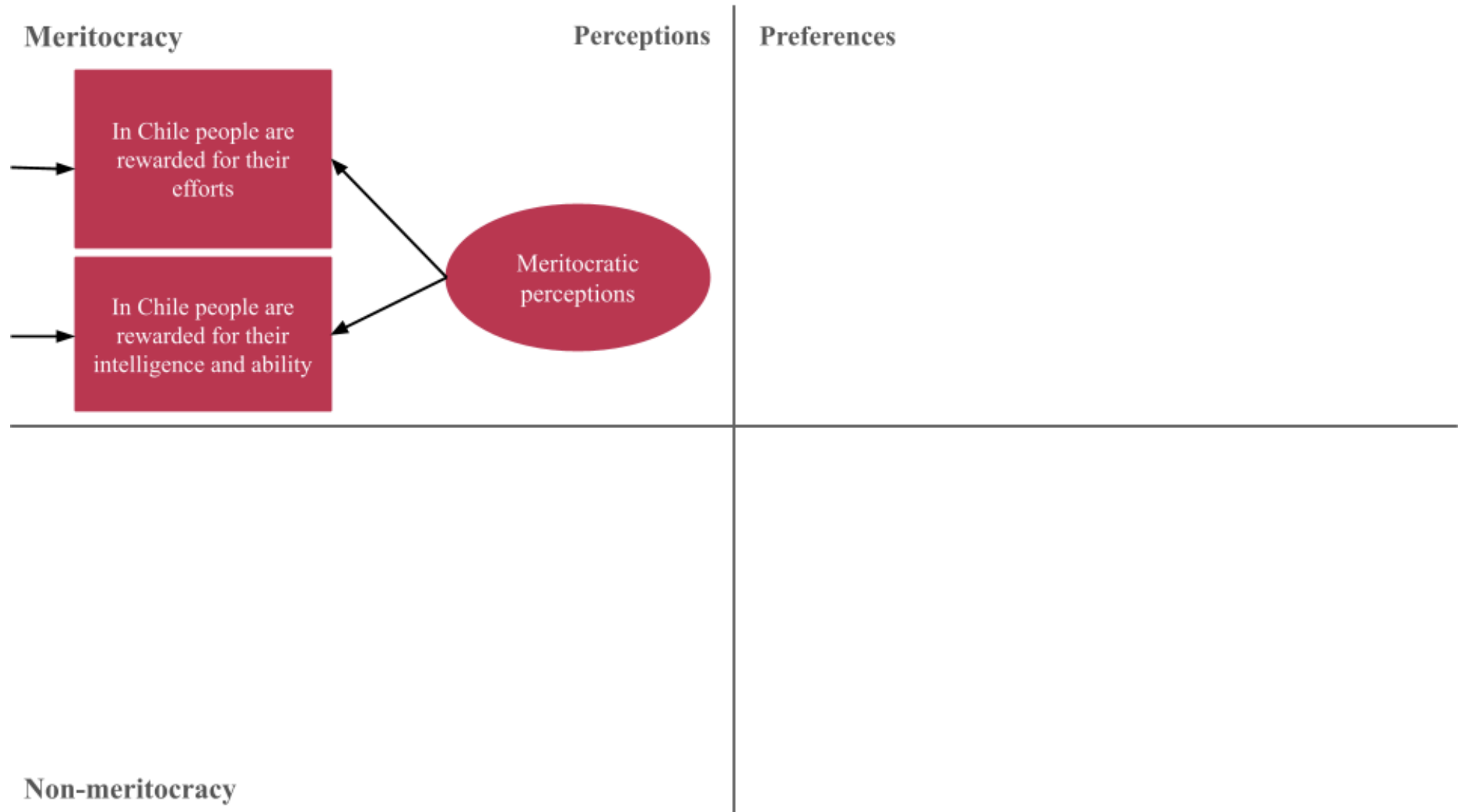
Data

- EDUMERCO: Online survey (CAWI), fielded in 2025, with adult respondents from the Metropolitan Region of Chile
- Non-probability sample with quota-based design
→ approximating population parameters by age, gender, education, and socioeconomic status
- Total sample: $N = 3,470$
- Analytical sample: $N = 2,557$

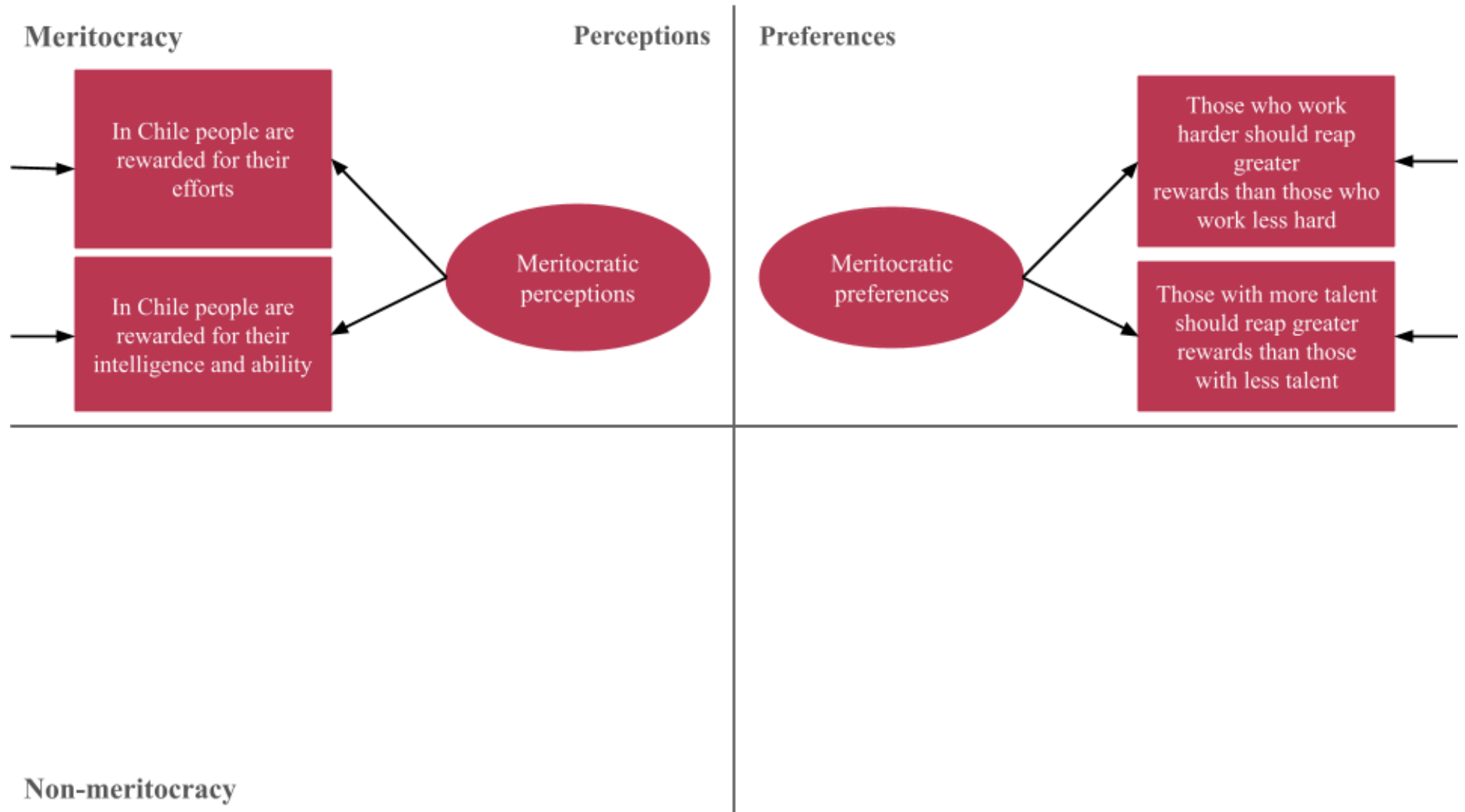
Measures



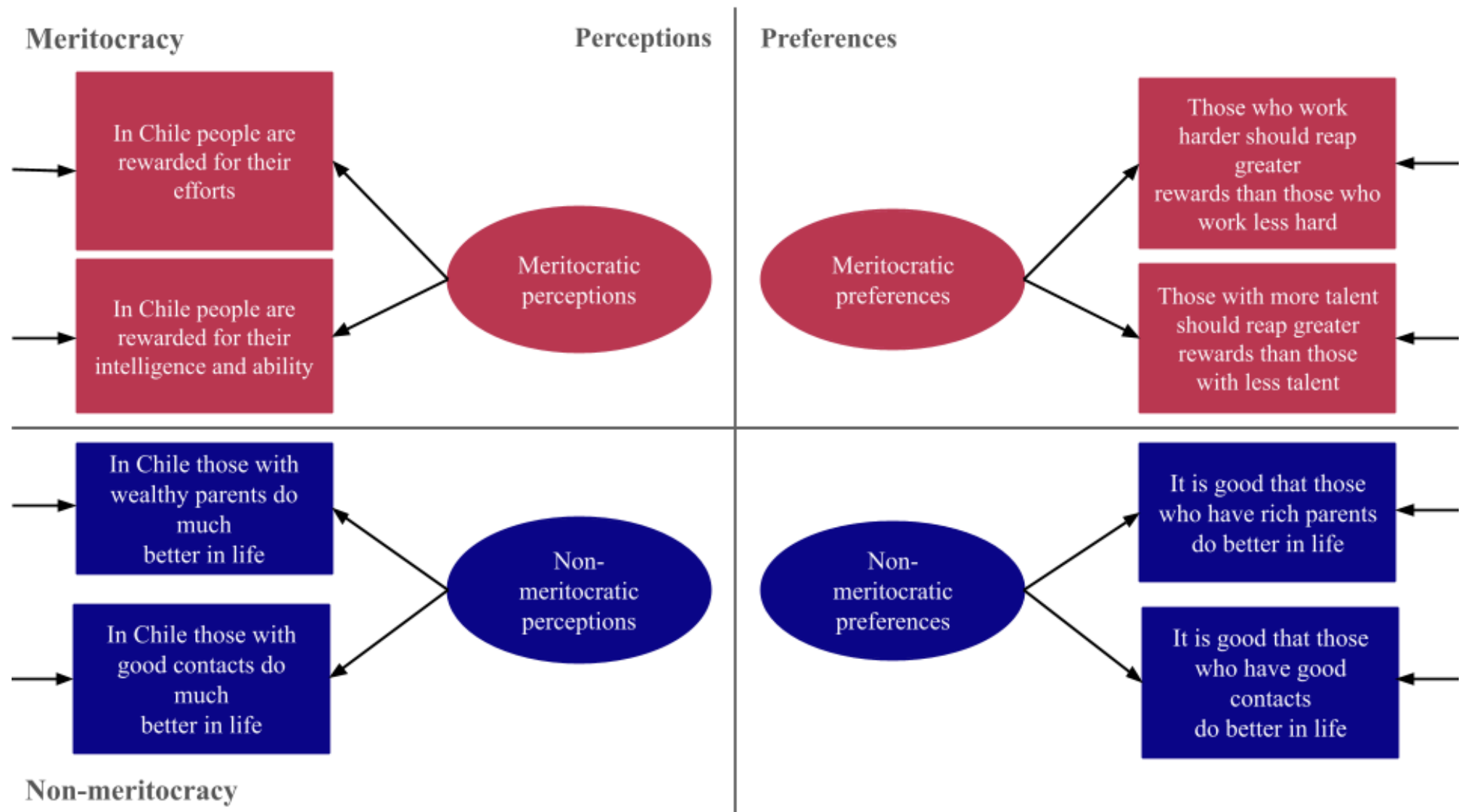
Measures



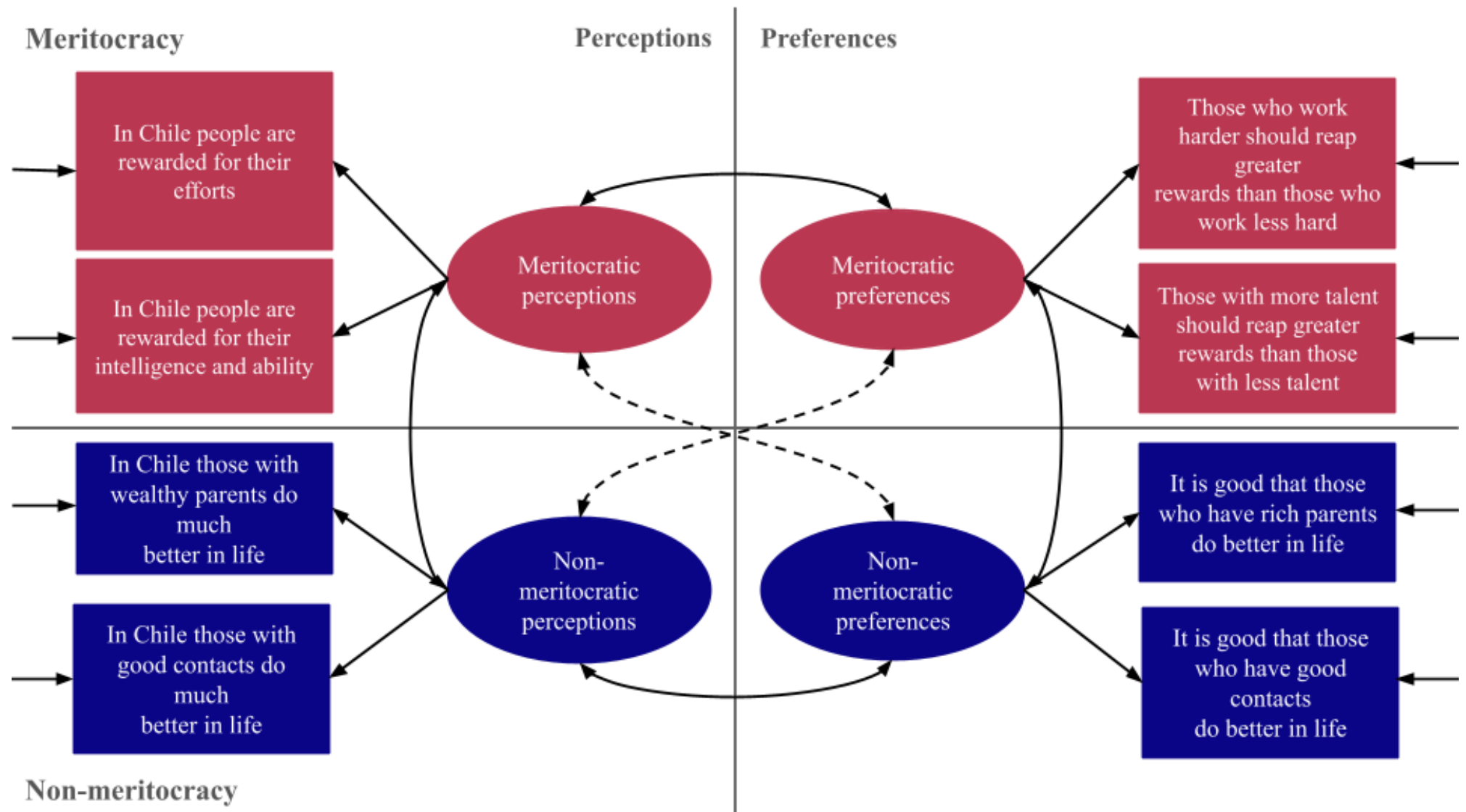
Measures



Measures



Measures



Analytical strategy

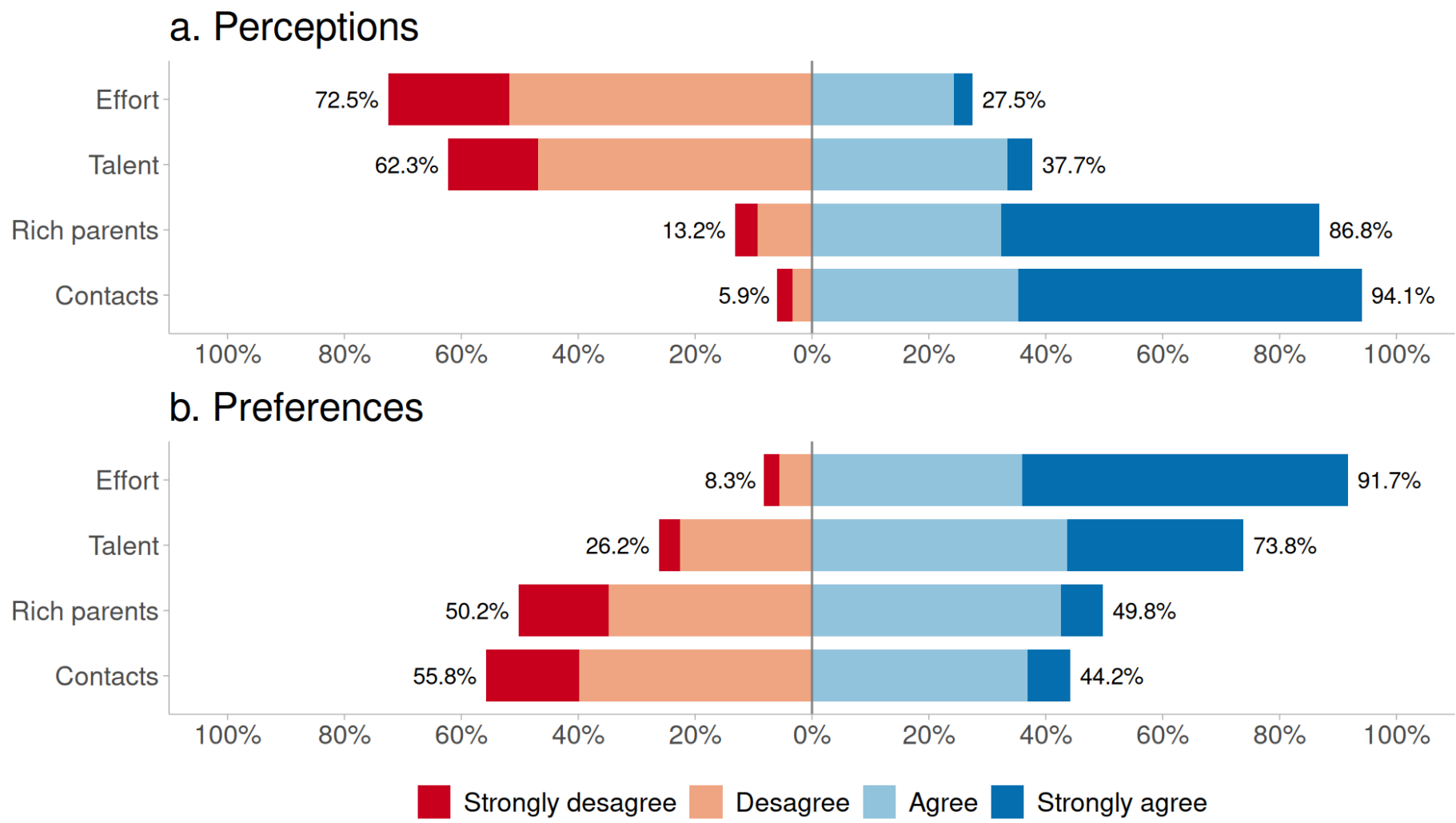
- First, we estimate a Confirmatory Factor Analysis of the multidimensional meritocracy scale
→ good model fit (CFI = 0.995, TLI = 0.991, RMSEA = 0.048)
- Second, we estimate a Structural Equation Model
→ pension market justice regressed on latent meritocracy factors

$$\text{MJP}_i = \alpha + \beta_1 \eta_i^{PM} + \beta_2 \eta_i^{PNM} + \beta_3 \eta_i^{PrM} + \beta_4 \eta_i^{PrNM} + \gamma' X_i + \varepsilon_i$$

- Estimation: WLSMV
→ appropriate for ordinal (Likert-type) data ([Kline, 2023](#))

Results

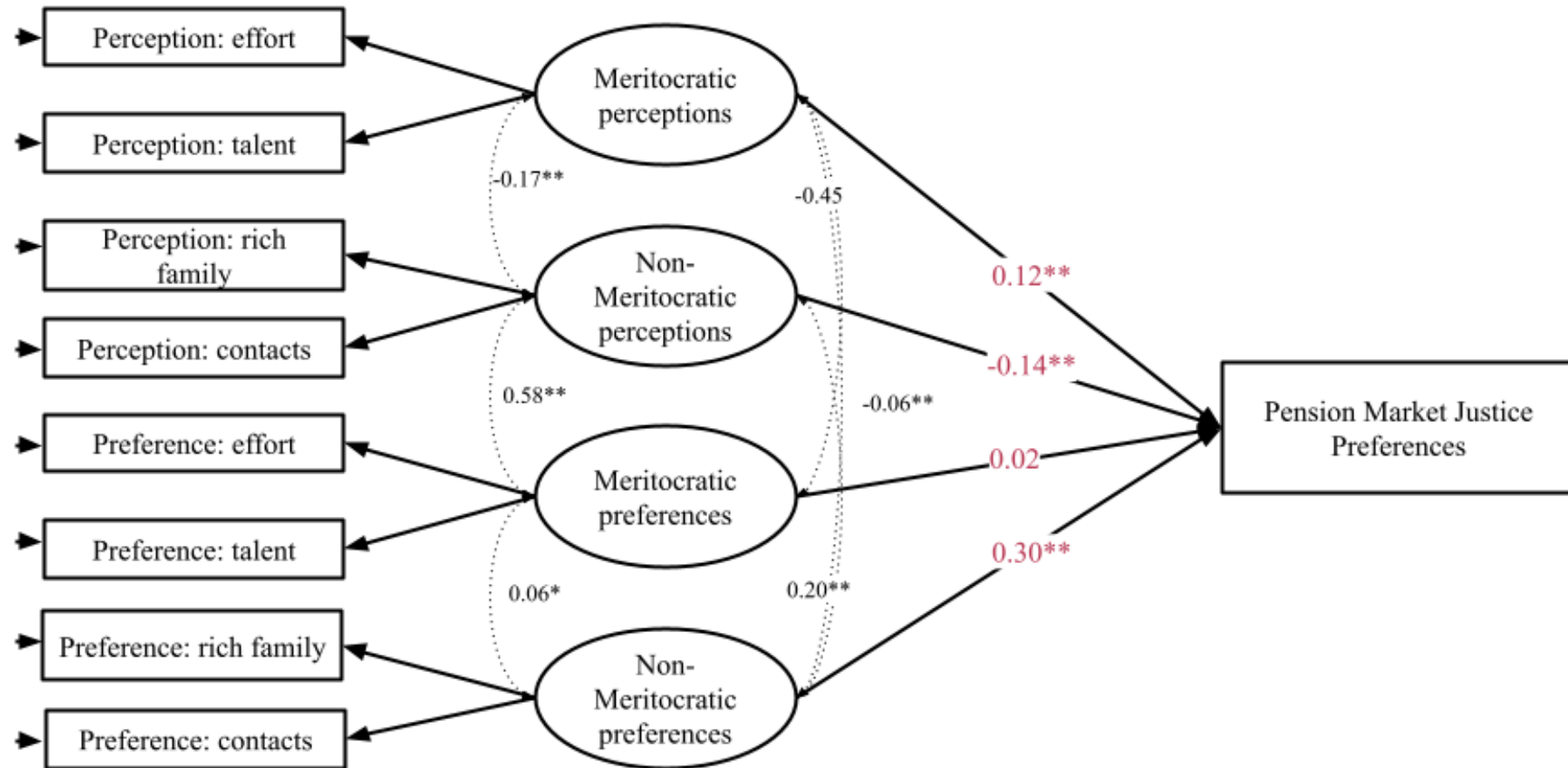
Descriptive



Source: own elaboration based on Survey EDUMERCO (n = 2557)

Figure 9: Distribution of responses on the scale of perceptions and preferences regarding meritocracy

Model



Model adjusted for: gender, age, income, education level, and political affiliation

WLSMV Estimation, standardized coefficients are shown, N=2,557
 Chi2=610.894*** / df=36, CFI=0.963, TLI=0.985, RMSEA=0.048
 ** $p < 0.01$, * $p < 0.05$

Figure 10: Structural equation model

Preliminary conclusions

1. **Meritocracy in Chile:** Privilege is perceived to outweigh merit, even though merit—especially effort—remains the dominant normative ideal
2. **Measuring meritocracy:** Respondents distinguish between perceptions and preferences, as well as between meritocratic and non-meritocratic principles
3. **Meritocracy and pension market justice:** Support is strongest among those willing to normalize privilege within a fair distributive order

Discussion and conclusions

Discussion and conclusions

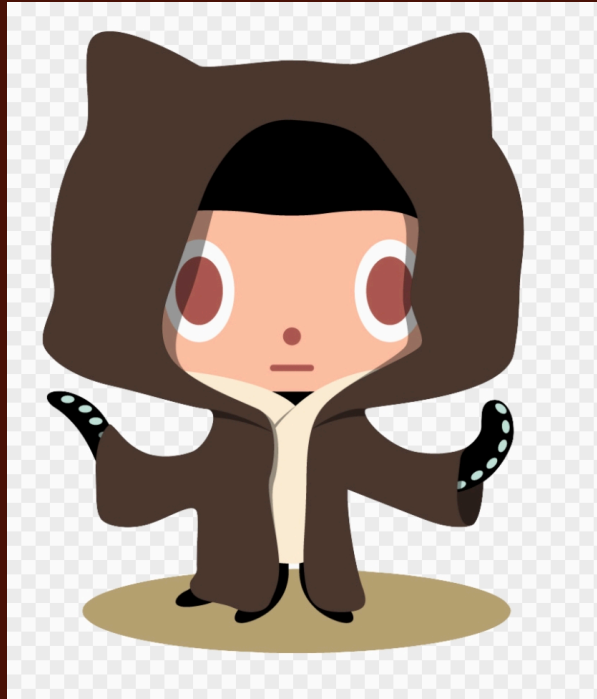
1. **Research agenda:** The expansion of market-based welfare is reflected in public attitudes ([Busemeyer & Iversen, 2020](#); [Lindh, 2015](#)), helping explain support for neoliberal arrangements ([Mau, 2015](#))
2. **Chile:** Support remains minority but is rising over time, consistent with longitudinal evidence from a highly commodified welfare context marked by mixed policy feedback ([Castillo et al., 2025](#))
3. **Main result:** Effort-based meritocratic perceptions are positively associated with pension market justice, both between and within individuals
4. **Extension:** With a multidimensional measure of meritocracy ([Castillo et al., 2023](#)), support is highest among those who normatively accept non-meritocratic advantage, while perceiving society as non-meritocratic lowers it

Next steps

- **Market justice measurement:** We are refining the measure and extending it to additional welfare domains
- **Class measurement:** Despite the null findings, we do not abandon the argument that objective stratification shapes these preferences; we will test alternative measures of class and inequality
- **Next analyses:** We will further examine how meritocratic deservingness relates to market justice in pensions, health, and education using latent class analysis and comparative ISSP data, with attention to institutional and economic cross-national differences

Thank you for your attention

- Github project: <https://github.com/jus-mer>



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Supplementary Material

Bivariate

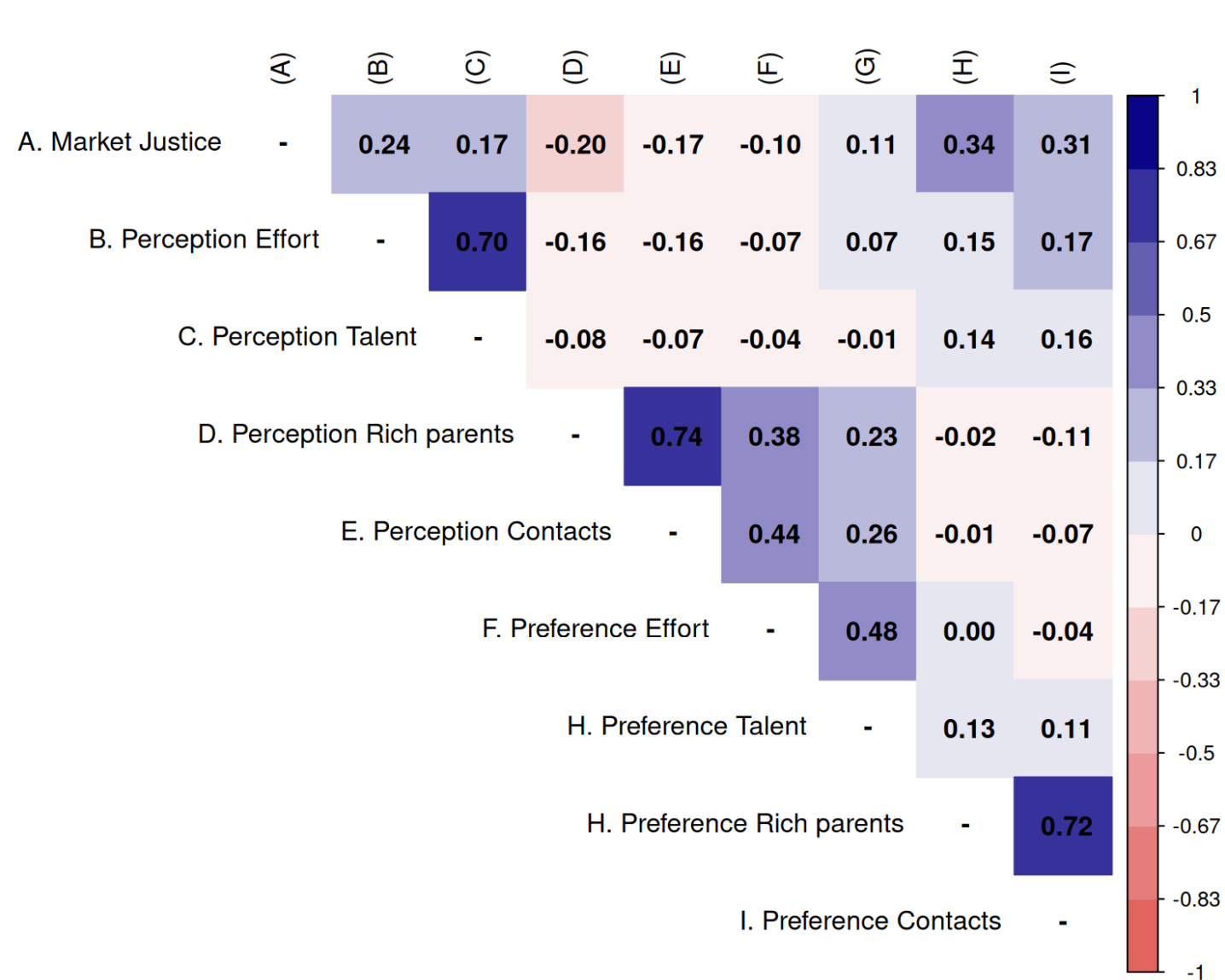
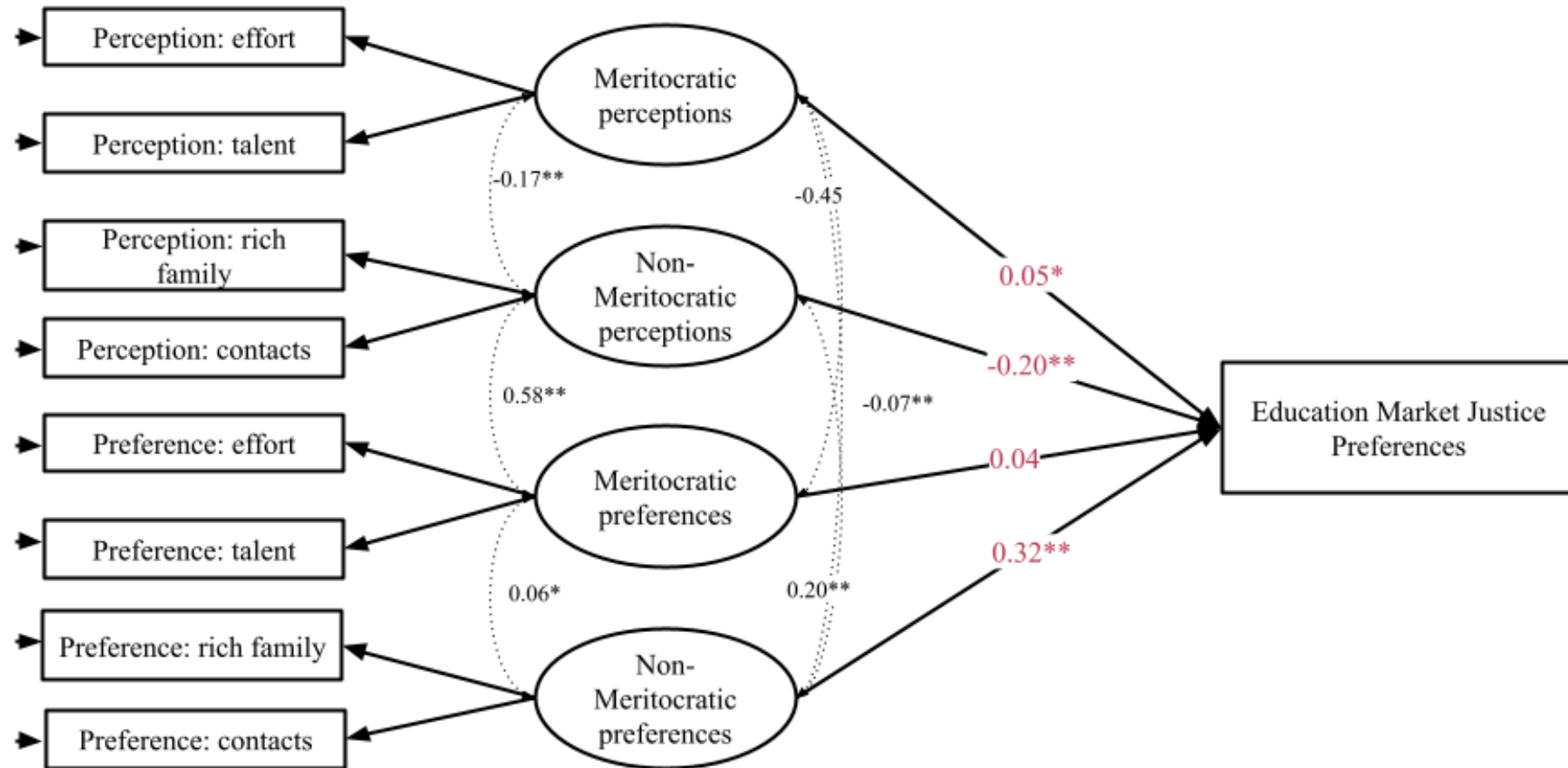


Figure 11: Correlation matrix between market justice and meritocracy scale

MIMIC Model: Education

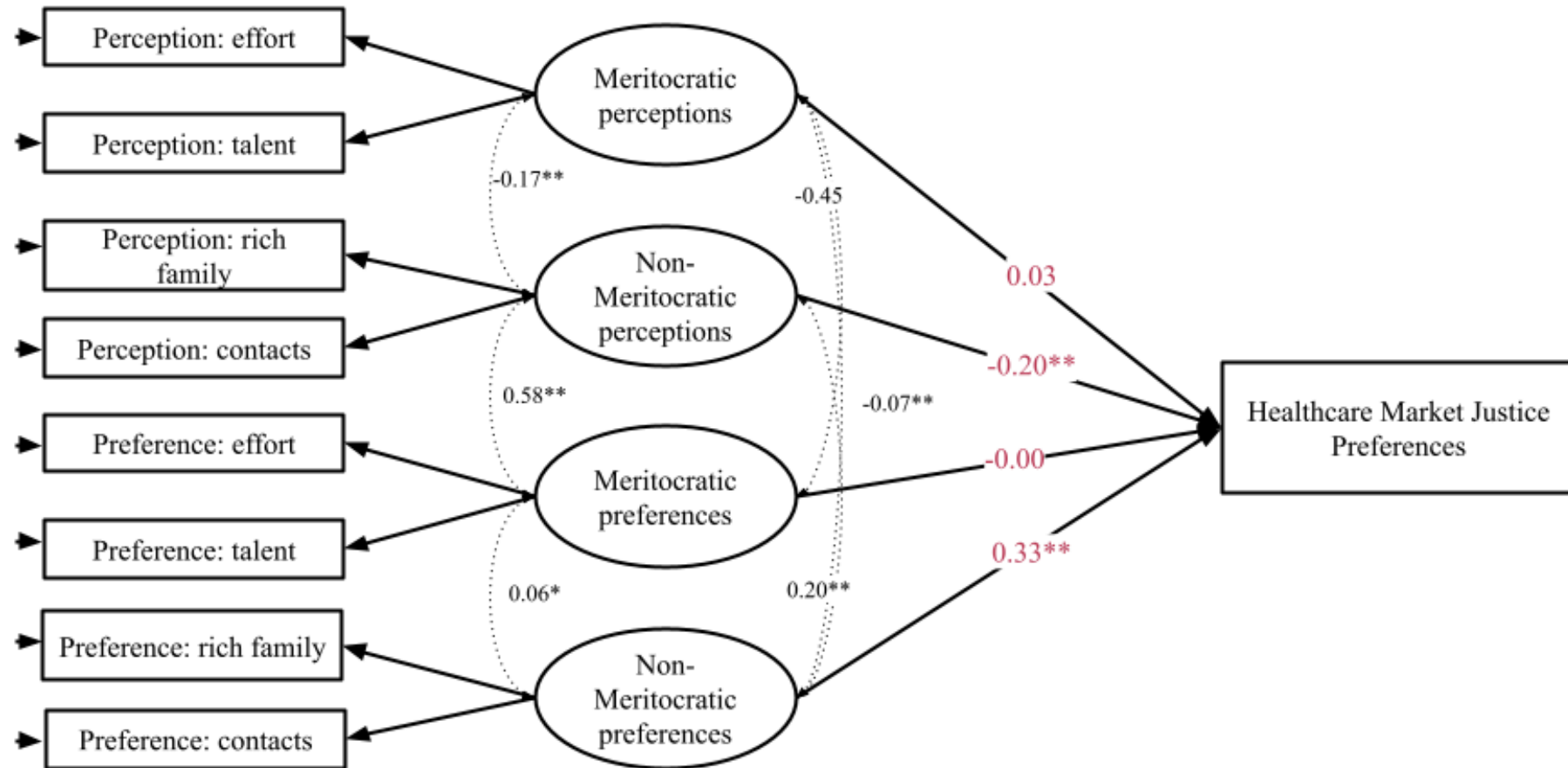


Model adjusted for: gender, age, income, education level, and political affiliation

WLSMV Estimation, standardized coefficients are shown, N=2,531
 Chi2=612.441*** / df=36, CFI=0.960, TLI=0.984, RMSEA=0.048
 ** $p < 0.01$, * $p < 0.05$

Figure 12: MIMIC Model for market justice preferences in education

MIMIC Model: Healthcare



Model adjusted for: gender, age, income, education level, and political affiliation

WLSMV Estimation, standardized coefficients are shown, N=2,540
Chi2=606.209*** / df=36, CFI=0.961, TLI=0.984, RMSEA=0.048
** $p < 0.01$, * $p < 0.05$

Figure 13: MIMIC Model for market justice preferences in healthcare

